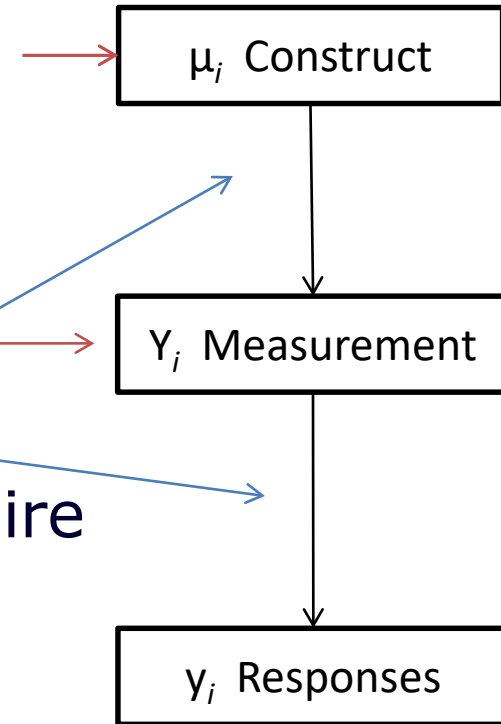


# Review: Overview of the questionnaire design process

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- Determine purpose/objective/topic
- Select mode
- Develop analysis plan
- Gather or write questions
- Pretest/test survey questions
- Organize questions into a questionnaire



# Cognitive Interviewing (1)

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“Cognitive interviewing techniques are used to study the manner in which targeted audiences **understand, mentally process, and respond** to the materials we present – with a special emphasis on potential **breakdowns** in this process.”

(Willis, 2005, p. 3)

Consider the specific questions that you have been working on, what are the potential breakdowns?

- Understanding:
- Mentally process:
- Respond:

# Cognitive Interviewing (2)

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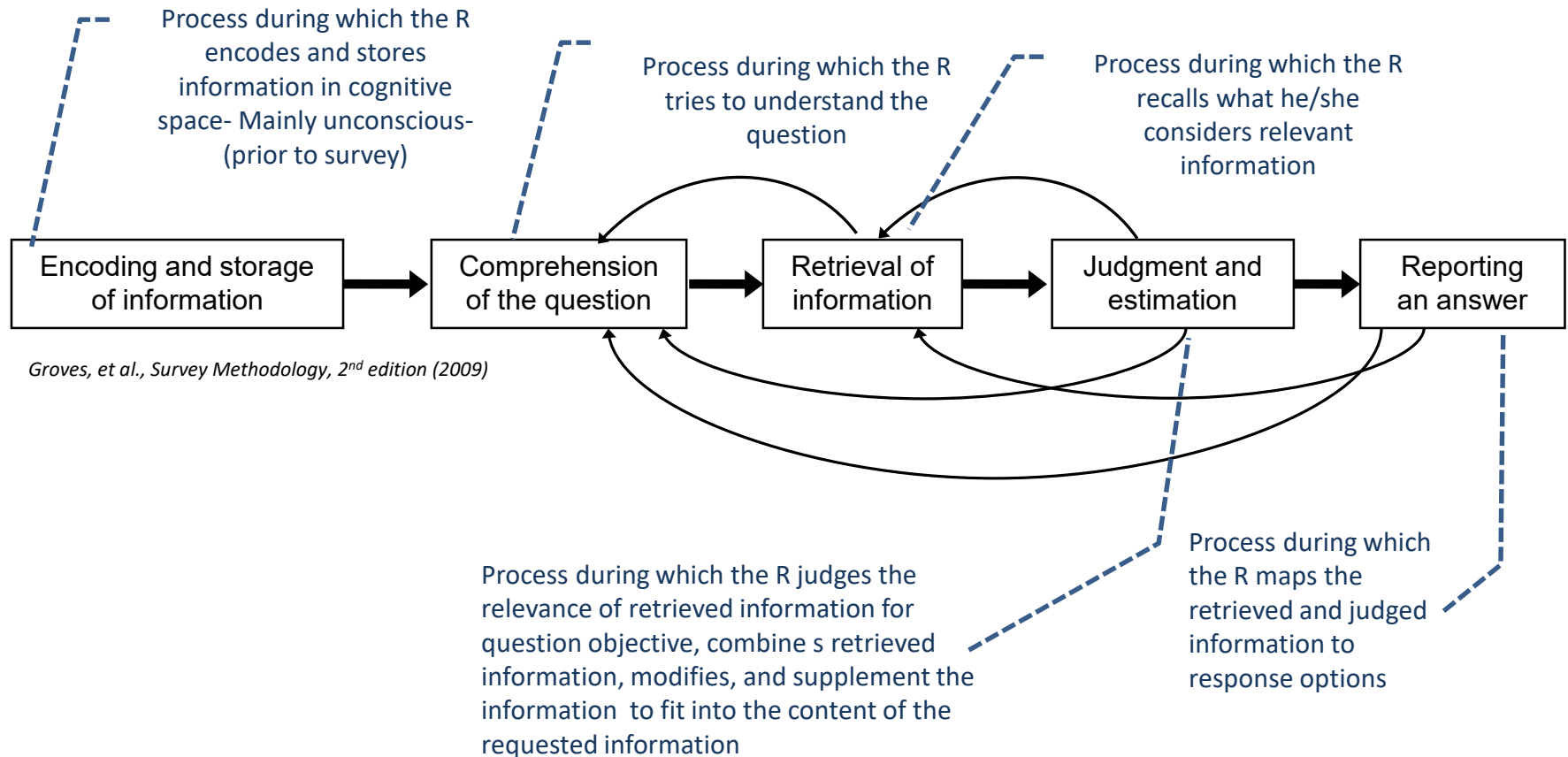
- Focuses on the cognitive processes that respondents use to answer a survey question
  - Model of survey response process as a guiding framework
- Identify problems
  - Related to *R*'s thinking (especially good for comprehension)
  - Also can learn about topic from *Rs* (who are experts) and uncover hidden assumptions
- Used to uncover:
  - What questions are problematic
  - Why they are problematic

# Question-Answering Process

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- Comprehension
  - Retrieval
  - Judgment
  - Response (Mapping and Editing)
- 
- NOTE: Respondents can go wrong and have difficulties at any of these stages.

# Respondent: Response Process



# What Can Go Wrong? (Or What Is A Problem/Breakdown)

---

- Comprehension
  - Question intent – What does  $R$  believe question asking?
  - Meaning of terms – What do specific words and phrases in the question mean to  $R$ ?
- Retrieval
  - Recallability of information - What types of information does  $R$  need to recall to answer question?
  - Recall strategy – What types of strategies using for retrieval (e.g., counting individually versus estimation strategy)
- Judgment
  - Estimation/judgment strategy
  - Motivation – Does  $R$  devote sufficient mental effort to answer accurately?

# What Can Go Wrong? (Or What Is A Problem/Breakdown)

---

- Response
  - Mapping - Can *R* match his or her internally generated answer to response categories given by question?
  - Editing - Does *R* want to tell the truth? Does he/she have something that makes him/her look “better”?
- Some processes conscious, some automatic
  - Automatic processes hard to detect with cognitive interviewing
  - *R*’s aware of conscious processes, but are they willing to report them?

# Write Good Questions That...

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- ...Rs can answer accurately
- ...Rs are willing to answer accurately
- Structure your cognitive interviews to identify issues related to both aspects



# Cognitive Interviewing: Theoretical Background (Ericsson & Simon, 1980, 1984)

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- Verbal protocols, protocol analysis
- Processes people use to solve different kinds of problems (proving theorems, playing chess, anagrams)
- Thinking out loud
- People can report what they are currently aware of (short term memory)
- **Assumption:** These reports are non-reactive—the act of thinking out loud does not change the process of answering or the response itself

# **Additional Tasks Built in Cognitive Interviewing (Presser et al., 2004)**

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- Think-aloud
- Verbal probing (concurrent/retrospective)
- Vignettes, card sorts, ratings tasks
- Response latencies

# Cognitive Interviewing: Implementation Issues

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*Develop a cognitive testing plan that covers:*

- Recruiting and scheduling subjects (time, \$\$)
  - How? Convenience sample, usually recruit through flyers, newspaper ads, community groups
  - What types? Representative of the population you are studying, make sure some respondents will go down all paths in questionnaire
  - Incentive usually given – \$25-\$50
- How many rounds of interviews and number of subjects per round? It depends, but
  - At least two rounds useful (3-4 helpful)
  - Generally, 5-15 subjects per round
  - Early rounds – conceptual issues, later – individual Qs

# Cognitive Interviewers

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- Typically experienced in questionnaire design (methodologist, cognitive psychologist) but not necessarily interviewing
- Should have technical skills (knowledge of questionnaire design) and ability to put respondent at ease and adapt to the unexpected
- Evidence that field interviewers may be used (especially with some training on how to uncover problems)
- Use multiple interviewers
- Experience with multiple interviewers conducting interview

# Training Cognitive Interviewers

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- How to train cognitive interviewers?
  - Trainees review questionnaires to identify problems and attend meetings to discuss questionnaire development
  - Learn background on cognitive interviewing technique and specific probing methods (including examples)
  - Observe experienced cognitive interviewers
  - Perform cognitive interview while being observed by experienced interviewer
  - Attend meetings to discuss cognitive interview findings
- Where conduct interviews? Cognitive labs (recording equipment) or natural settings (quiet)
- Tape (video) record interview and take notes on problems
- Mode? In-person may be preferred, but telephone works too (More options: Skype, Second Life)
- How long of an interview? 15 min-2 hr, 1 hour recommended (20-30 minute questionnaire)

*Now...Develop cognitive protocol*

# Cognitive Interviewing Protocol

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- The **Protocol** is a guide which includes *questions* and *scripted probes developed before interview* (include white space to take notes)
  - Be flexible
  - Add probes
  - Follow up on information provided by respondent
- Combination of (1) think aloud (probes to encourage) and (2) debriefing/directed probes
- How scripted to be?

# Think-Aloud

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- In responding to a question, the respondent is asked to both answer the question and verbalize his/her thoughts. The verbal think-aloud stream is analyzed to determine the processes the respondent used to construct an answer to the question.
- Assumes...
  - Reports on thought processes available to *R*
  - *Rs* can report these thought processes
  - Reporting process does not change the thought process

# Example

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**Question:** How many times have you talked to a doctor in the last 12 months?

**Response:** “I guess that depends on what you mean when you say ‘talked.’ I talked to my neighbor, who is a doctor...I go to my doctor about once a year for a general checkup...I’ve also probably been to some type of specialist [bad knee, coughing]...which I’m pretty sure was in the past year...I’ve also talked to doctors...when I brought my kids in to the pediatrician...assume you don’t want that included...saw a chiropractor...I don’t know if that’s a doctor...”

(Willis, 2005)



# More About Think-Aloud

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- “Train” the respondents to think out loud before the cognitive interview starts
  - “Try to visualize the place where you live, and think about how many windows there are in that place. As you count up the windows, tell me what you are seeing and thinking about.” (Willis, 2005)
  - May need to do another example to get *Rs* comfortable
- Include some probes especially early on to help *Rs* get used to the process (“*Tell me more about that*”, “*Tell me what you are thinking*”)
- Typically done concurrently (as *Rs* are answering question)
- May be difficult for some *Rs*

# Cognitive Probes (1)

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- Concurrent probing
  - Interviewer asks the survey question
  - Respondent answers the question
  - Interviewer asks a probe question
  - Respondent answers the probe question
- Retrospective probing
  - Interviewer asks (and respondent answers) all the survey questions
  - Interviewer asks (and respondent answers) probe questions

# Cognitive Probes (2)

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- Proactive probes
  - Researcher actively probing to search for problems
  - What to probe may be specified in advance of interview
  - Scripted in advance (anticipated) or left to interviewer to formulate (spontaneous)
- Reactive probes
  - Based on respondent answers that suggest a problem
  - Standard probes from a prepared set can be presented as needed
  - Probes can be scripted in advance (conditional) or on the spot (emergent)

# Cognitive probes (3)

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- Scripted probes
  - Scripted probes built into cognitive interviews
    - Good for inexperienced interviewers,
    - Identifying areas more likely to be problematic
- Spontaneous probes
  - Probes on the spot
    - Cannot anticipate all problems will encounter, allows for flexibility
  - Good for experienced interviewers only

# Example Probes

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- Comprehension/Interpretation
  - *What does the term outpatient mean to you?*
  - *Can you repeat the question I just asked to you?*
- Recall Probe
  - *How do you remember that you went to the doctor five times in the last 12 month?*
- Judgment/Estimation
  - *How did you arrive at this answer?*
  - *How sure are you that your health insurance covers drug treatment?*
- Mapping/editing
  - *How personal do you think this question is to people like you?*

# Example Probes (2)

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- Proactive probes
  - *Remember to tell me what you are thinking.*
  - *What makes you think that?*
  - *What were you thinking about as you were trying to remember that?*
  - *What types of events did you include in your answer?*
- Reactive probes
  - *You look puzzled, what are you thinking?*
  - *You answered that very quickly. Why was that it easy for you to come up with your answer?*
  - *I appreciate the effort you're making to recall this information.*

# Let's Try This Out Now...

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- *During a typical work day at your job as a survey methodologist, how much time do you spend doing strenuous physical activities such as lifting, pushing or pulling? Please note that complex variance estimation and pulling your hair out after working with a difficult client do not count as strenuous physical activities.*
  - \_\_\_ *None*
  - \_\_\_ *Less than 1 hr*
  - \_\_\_ *1-4 hours*
  - \_\_\_ *4+ hours*
- What kinds of problems?
- How would you probe?
- What do you think was learned and how was question modified?

# Cognitive Interviews: Think-aloud

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- In responding to a question, *R* is asked both to answer *Q* and to verbalize his/her thoughts
- Verbal think-aloud stream is analyzed to determine processes respondent used to construct answer to *Q*
- Example
  - *I'wer*: "How many times have you talked to a doctor in the last 12 months?"
  - *R*: "I guess that depends on what you mean when you say 'talked.' I talked to my neighbor, who is a doctor...I go to my doctor about once a year for a general checkup...I've also probably been to some type of specialist [bad knee, coughing]...which I'm pretty sure was in the past year...I've also talked to doctors...when I brought my kids in to the pediatrician...assume you don't want that included...saw a chiropractor...I don't know if that's a doctor..."



# Cognitive Interviews: Probes

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- Retrospective protocol: *How did you arrive at that answer?*
- Paraphrase: *Can you tell me in your own words what that last question was asking?*
- Confidence ratings: *How sure are you about your answer to the last question?*

# Cognitive Interviews: Example 1

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- *I'wer: "For each problem listed below, please indicate to me whether it is a very urgent problem, a somewhat urgent problem, a small problem, or not a problem at all in your neighborhood. Drug abuse."*
- *R: "...Mm."*
- *I'wer: "Tell me what you're thinking."*
- *R: "Well, I haven't, heard of, any, well. I haven't heard of any, drug use in my neighborhood."*
- *I'wer: "Okay."*
- *R: "Um....So, um...In my neighborhood. Um, so, um...I don't know how to answer that. I don't, know."*

# Cognitive Interviews: Example 1

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- *I'wer: "Why don't you know how to answer it?"*
- *R: "Just, because since, I don't, hear of ANY, I don't know like, what answer you're looking for, because..."*
- *I'wer: "What, whatever it means to you. Again, the question, for each problem listed below, please indicate to me whether it is a very urgent problem, a somewhat urgent problem, a small problem, or not a problem at all in your neighborhood and the first one..."*
- *R: "Oh, it's a very urgent problem."*
- *I'wer: "Okay."*

# Cognitive Interviews: Example 2

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- *I'wer: "Transportation of high-level radioactive wastes, means that state and local governments may have to spend money on things like emergency preparedness. Do you think that the federal government should reimburse the states and localities for costs like these?"*
- *R: "Um...I don't want to say no, because I think ultimately states...should handle their own business, unless it's a federal issue..."*
- *I'wer: "I noticed you paused there for a second after I asked you that. Was there something unclear in that question?"*
- *R: "Um...yeah...I would, unless I wasn't hearing the question, completely clearly, I...I wasn't sure WHO was producing the waste. Which that may have been the case I was thinking back to what the question asked, what...you know who...who was producing the waste whether it was federal government or state."*
- *I'wer: "Alright."*

# Cognitive Interviews: Advantages

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- Also relatively fast and inexpensive – Can do 10 to 15 cognitive interviews in a few weeks for a few thousand \$\$
- Can revise and test new versions of Qs
- Often reveals comprehension or memory problems that other methods would not find
- Effective means for persuading sponsors their favorite items have problems

# Cognitive Interviews: Drawbacks

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- Data reduction/summary can be difficult and subjective
- Inconsistent interpretation of whether or not a problem has been expressed
  - Conrad & Blair (2009) asked coders to listen to interviewer transcript and found overall inter-coder agreement low
    - Presence of problem: average kappa = .31
    - Type of problem: average kappa = .39
    - Especially when no evidence of a problem prior to probe
- Likely to produce false alarms: *I'wers* believe is problem, coders do not

# Cognitive Interviews: Drawbacks

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- Method not really standardized; different techniques used by different organizations, *I'wers*
  - Beatty (2004) found that *I'wers* differed in frequency with which they used “confirmatory probes” (e.g., mirroring back what *R* said)
- No way to judge how representative findings are, how common problems are
- Inconsistent solutions to identified problems
  - DeMaio & Landreth (2004) found that different approaches to cognitive interviewing led to different solutions that were differentially effective

# Cognitive Interviews: Drawbacks

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- Limits on introspective access
  - People can only report conscious contents of working memory
  - May report theory about process rather than process itself
- Effect on data quality largely unknown
  - Fowler (2004) demonstrated with split ballot study that reporting seems to be improved when questions revised after cognitive interviewing



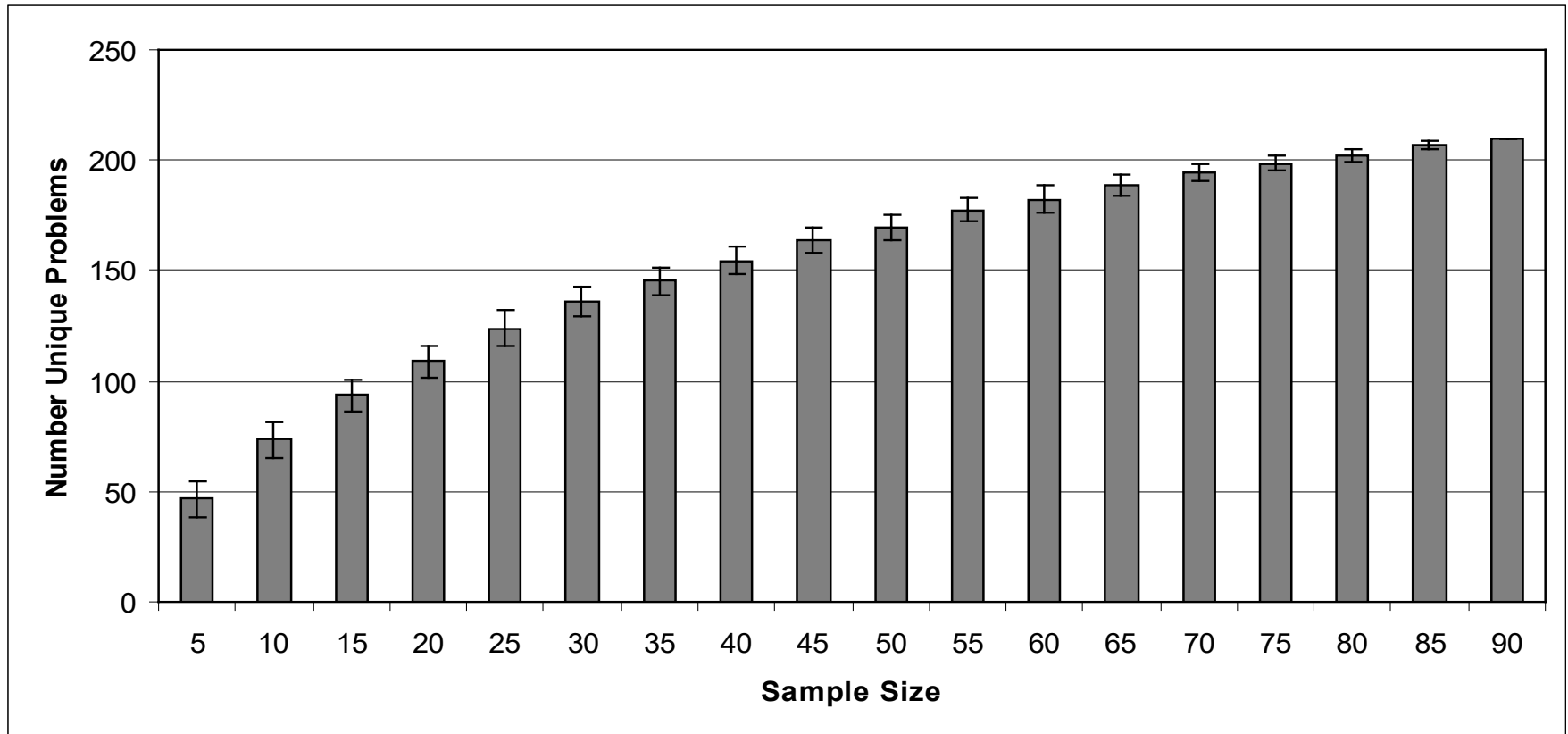
# Cognitive Interviews: Drawbacks

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- Small samples may not be sufficient to detect many problems; including the most serious problems
  - Blair, Conrad, Ackerman & Claxton (2006) and Blair Conrad (2011) conducted 90 cognitive interviews
    - Selected 90 samples, with replacement, of different sizes: 5, 10, 15, ... 90
    - Compared problem detection for each sample size
    - 3 experts in questionnaire design and cognitive interviewing rated each coded problem for its “impact” on data quality
    - Are most problems detected in small samples?
    - Are most serious (high impact) problems found in small samples?

# Cognitive Interviews: Drawbacks

- Mean number of unique problems at each sample size



Source: Blair & Conrad (2011)

# Drawbacks - Summary

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- Better at diagnosing problems than finding solutions
- Small, unrepresentative samples of paid subjects (may miss problems that would occur with larger, diverse population)
- Different problems may occur in more real world field settings
- Different interviewers may produce different conclusions
- Respondents unable to report on some processes at all or report inaccurately on process

# Administering Cognitive Interviews

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- Get recording set up
- Introduction and obtain consent
- Use protocol with scripted probes as a guide
  - Remember: Sometimes there will be no process, and *Rs* will have nothing to report (e.g., date of birth) – they “just know”
- Thank *R* and answer questions

# Analysis of Cognitive Interviews

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- Write-up of individual interviews
- Enter and analyze quantitative data from interviews
- Debriefing of cognitive interviewers
- Summary report of findings, which is based on:
  - General qualitative assessment (most common)
  - Coding of respondent-interviewer interactions
  - Transcription, coding, qualitative analysis
- Questionnaire revised (and another round)
- What to look for?
  - Problems common across interviews, “discoveries” - problems may seriously threaten data quality or may be frequent in main study
  - Be careful to know limitations – may have idiosyncratic interview or know that your subjects are too highly educated relative to population

# One Summary Report Format

(Willis, 2005)

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- Description of general purpose of testing
- Methodology (subject population, recruitment, # of subjects, # of interviewers, specific procedures used)
- Protocol as an appendix (or include probes in the question-by-question summary described below)
- Provide overall written summary of most significant problems, prior to detailed question-by-question listing

# One Summary Report Format (2)

## (Willis 2005)

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- Question-by-question listing of problems and possible solutions to address them:
- Example from the first draft of the questionnaires

# Methodological Research/Future Direction

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- Reliability of findings from cognitive interviews (DeMaio & Landreth, 2004)
- Evidence of improvement of data quality (Fowler, 2004)
- Determining optimal sample size for cognitive interviews (Beatty & Willis, 2007; Blair & Conrad, 2011)
- How to conduct them
  - Balance between think-aloud & probes
  - Probing choice by type of question, mode, participant
- Interviewer effects
- Better standardized documentation



# Write good questions that

---

- Rs can answer accurately
- Rs are willing to answer accurately

# Feedback on Cognitive Interview Probes

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- Plan for 30 minutes
- Strategically spend more time on the more important questions
- Think about all the stages and possible sources of error, and make sure that your probes cover the suspected areas
  - Comprehension: Sentence structures, ambiguous words, etc.
  - Retrieval and Judgment: Attitude/strategy towards uncertainty, knowledge
  - Mapping and Editing: Response categories, social desirability bias
- Format the protocol to capture the responses and notes
- Don't rely on R's memory as you move from one question to another
- Capture starting and ending times

# Cognitive Interview Implementation (1)

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- Complete 3 cognitive interviews with acquaintances as respondents
  - Each interview should be about 30 min
  - Get recording set up (if possible)
- Read introduction and obtain consent
- Use protocol with scripted probes as a guide
- Take notes on the *Q'naire* and record (if possible)
- Thank *R*, and answer any questions that they have

# Cognitive Interview Implementation (2)

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- Written summary report of findings
  - Who were your respondents?
    - Example: *R1 is a 35 year old married male with an associate's degree who has lived in Belleville, Michigan all of his life. He works as a manager at a grocery store. R1 is very articulate.*
  - General observations
    - Example: *R1 seemed very interested in the topic and had little difficulty with the questions. R2 had not thought much about these issues. Both respondents had difficulty with the questions about [TOPIC]. Both Rs had situations that did not fit the questions on [TOPIC] very well.*
  - For each section of questions
    - Indicate what probe you used
      - P1, P2, P3, ...
      - If you added your own probe, what was it?
    - Identify/report key problems
    - Possible suggestions for how they might be addressed
- Submit written report (Canvas)
- Make sure that you dispose physical *Q'aires* properly